

MODEL CONTEXT PROTOCOL

G R O U P M



PRESENTERS

Bibika Ghimire, Mateusz Nowak, Muhammad Sohaib, Micheal McCabe, Summer Rhoda



WHAT IS MCP

The Model Context Protocol (MCP) is an emerging open standard that defines a unified, bidirectional communication and dynamic discovery protocol introduced by Anthropic. It is designed to connect AI models with external systems through a universal protocol.

- It provides a consistent way for AI to access data, services, and tools securely and efficiently..

MCP eliminates the need for custom integrations for each AI tool combination. between AI models and external tools or resources, aiming to enhance interoperability and reduce fragmentation across diverse systems.



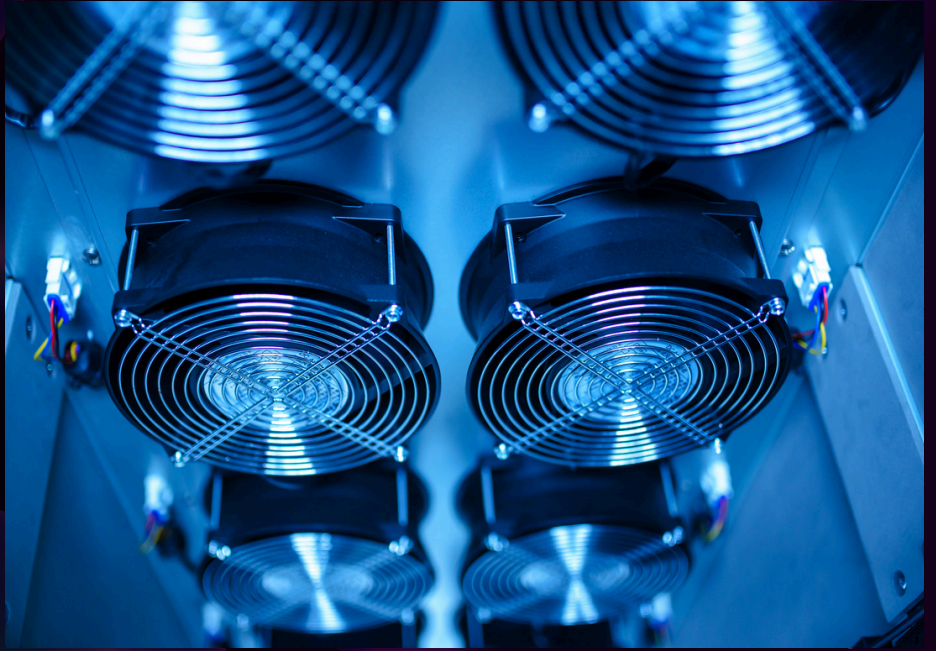
INTRODUCTION TO MCP!

In recent years, the vision of autonomous AI agents capable of interacting with various tools and data sources has gained significant momentum. This progress accelerated in 2023 with the introduction of function calling by OpenAI, which allowed language models to invoke external APIs in a structured way. This advancement expanded the capabilities of LLMs, enabling them to retrieve real-time data, perform computations, and interact with external systems. As function calling gained adoption, an ecosystem formed around it. OpenAI introduced the ChatGPT plugin, allowing developers to build callable tools for ChatGPT. LLM app stores such as Coze and Yuanqi have launched their plugin stores, supporting tools specifically designed for their platforms. Frameworks like LangChain and LlamaIndex provided standardised tool interfaces, making it easier to integrate LLMs with external services. Other AI providers, including Anthropic, Google, and Meta, introduced similar mechanisms, further driving adoption. Despite these advancements, integrating tools remains fragmented.



SOHAIB

HOW MCP SERVER WORKS



HOW MCP SERVER WORKS

- MCP (Model Context Protocol) is a system that lets AI models (like ChatGPT) connect to external data or tools safely.
- The MCP server acts as a bridge — it provides structured information or actions that the AI can use.
- The AI (client) sends requests → the MCP server processes them → and sends back results.
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- It helps the AI stay updated, secure, and extendable without direct internet access.

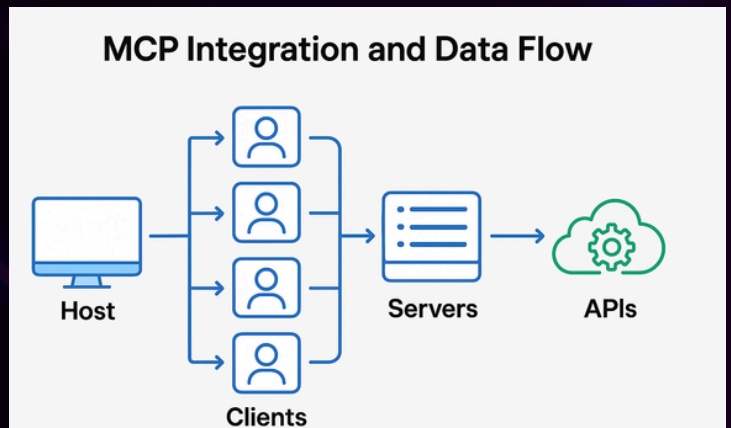


MCP Integration and Data Flow

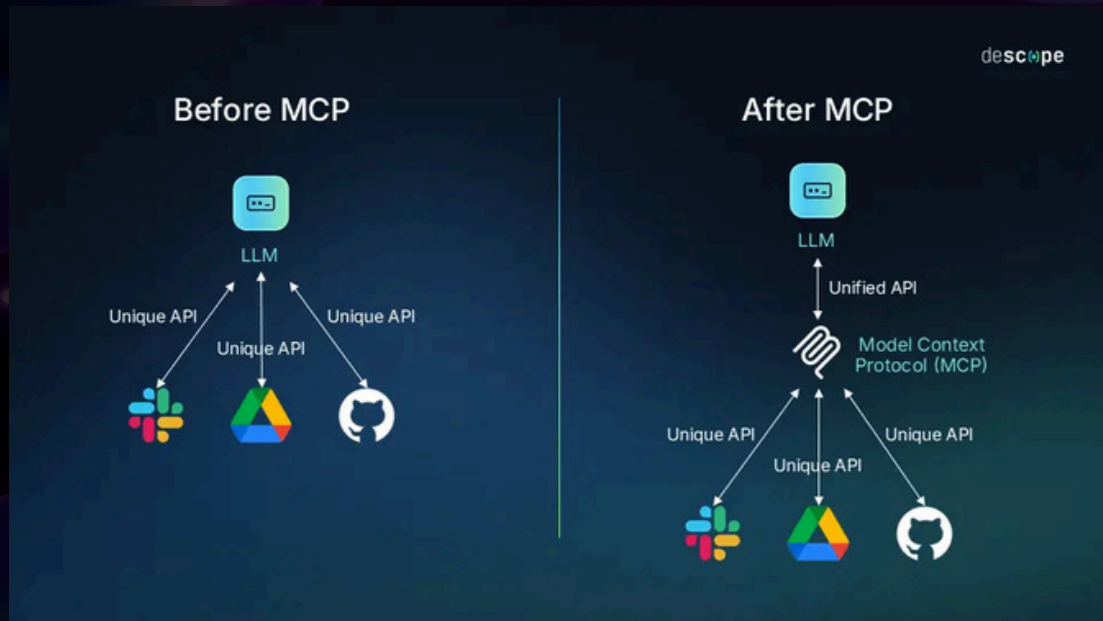
- **Host:** The main environment where MCP clients run (e.g., AI system or application).
- **Clients:** Individual units that send requests to MCP servers; each maintains its own connection.
- **Servers:** Receive requests from clients, process them, and provide responses or data.
- **APIs:** External data sources that MCP servers connect to for real-time information.

Data Flow:

1. Host runs multiple clients.
2. Clients send independent requests to MCP servers.
3. Servers gather or process data—often via APIs.
4. Responses flow back to clients → host → AI system.
5. Enables parallel communication, data scalability, and modular integration across systems.

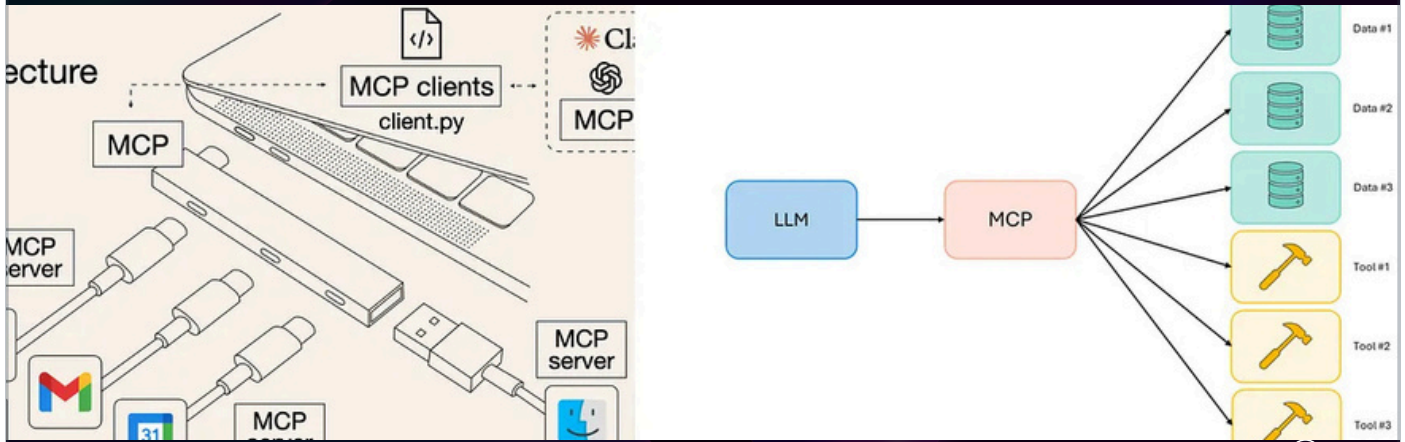


Before & After MCP



MCP Integration and Data Flow

- MCP Servers are essentially a standardized API for accessing your tools or data.



MCP Architecture and Core Components

1. MCP Host (AI Application)

The front-end AI agent or application (like Claude Desktop or IDEs) that the end-user interacts with

The host initiates connections to one or more MCP servers

2. MCP Client

A connector component running within the host application that manages the communication with a single MCP server

The host will spawn an MCP client for each server it wants to use, each maintaining a one-to-one connection

This design ensures isolation and security between integrations

3. MCP Server

A lightweight service (which can run locally or remotely) that implements the MCP protocol and offers a specific set of capabilities to the client

An MCP server could expose a database, a content repository, a SaaS application's API, or other data source

4. RPC (remote procedure call) Protocol

Communication between MCP clients and servers is handled by this layer

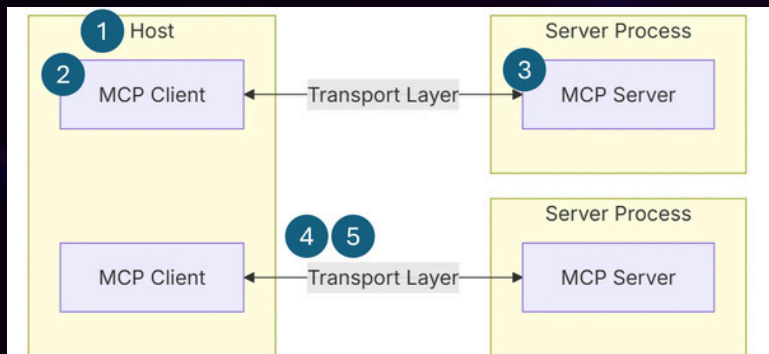
This layer is built on JSON-RPC 2.0, a well-established remote procedure call format

All messages (requests, responses, errors, notifications) conform to a common JSON structure, enabling interoperability across languages and platforms

5. Transport

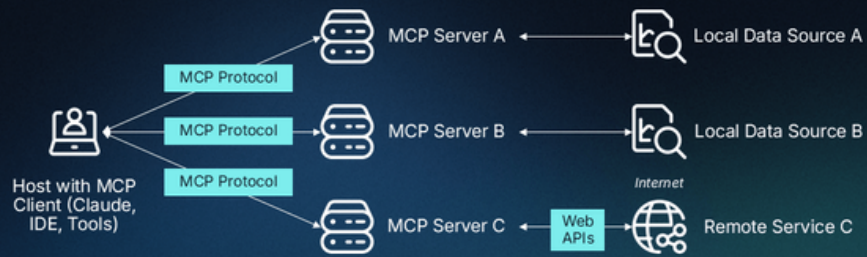
MCP supports multiple transport mechanisms for these JSON-RPC messages.

For servers running locally: STDIO (standard input output)
For remote servers:

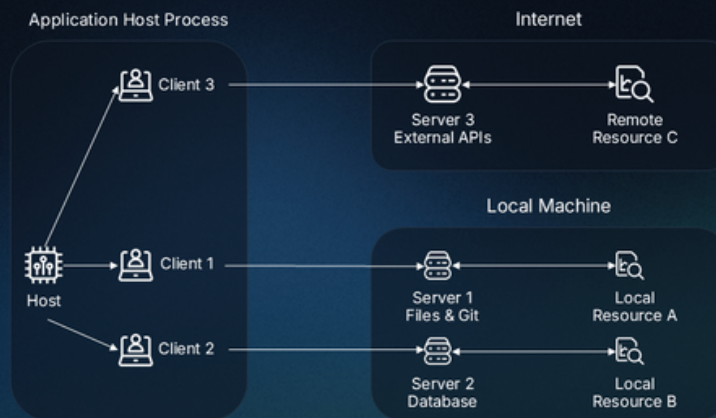


MCP Architecture and Core Components

- MCP Architecture



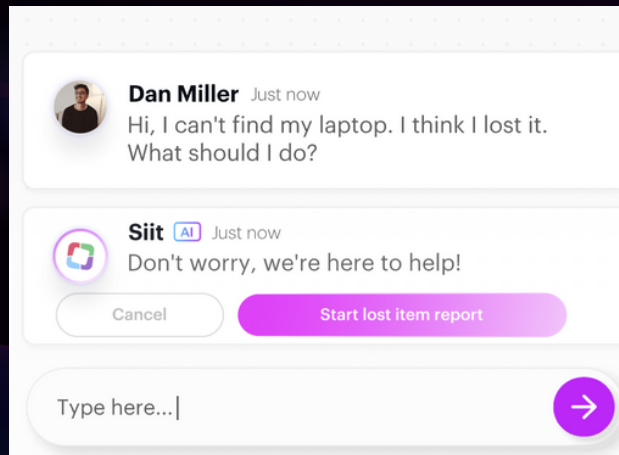
- MCP Components



Real Life Examples

Your paragraph text

- Siit, a modern ITSM platform, powers an IT AI agent that supports this use case



The screenshot shows a chat window with a white background and a light gray border. At the top, there's a header bar with a series of small dots. The chat history shows two messages: a user message from 'Dan Miller' (with a profile picture) asking for help finding a lost laptop, and a response from 'Siit AI' (with a colorful circular logo) offering assistance. Below the AI response are two buttons: 'Cancel' and 'Start lost item report'. At the bottom of the chat window is a text input field with the placeholder 'Type here...' and a purple circular button with a white right-pointing arrow.

Dan Miller Just now
Hi, I can't find my laptop. I think I lost it. What should I do?

Siit AI Just now
Don't worry, we're here to help!

Cancel Start lost item report

Type here... | ➔



Real Life Examples

Your paragraph text

- Ramp's AI agents can analyze individual transactions on finance team's behalf



Agent approved



Valid receipt



Under \$75



In-policy



Real Life Examples

Your paragraph text

- Juicebox's AI Recruiting Agents can support this use case

✦ I'm looking for senior engineers in the Bay Area who have built search infrastructure in the past

... Looking for profiles that match your criteria...

... Analyzing and learning from your feedback...

⚙ We have 35 profiles are ready for your approval



Real Life Examples

Your paragraph text

- Aleph AI offers an army of AI agents that can perform a host of actions on user's accounting data

✦ AI Mapping

Accept all Reject all

➤

Generate a mapping that rolls accounts up into one of the following categories: "Revenue", "Cost of Revenue", "Operating Expenses, or "Other"

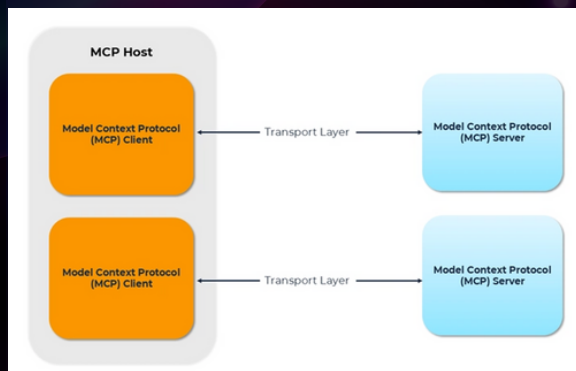
Continuous suggestions

Account	→	Roll-up group	Status
Product Sales		Revenue	To review
Hosting Costs		Cost of Revenue	To review
Service Revenue		Revenue	Accepted
Salaries & Wages		Operating Expenses	Accepted



USE CASES IN REAL LIFE





IN ENTERPRISE

- With MCP, AI assistants can use data that's shared between various departments in a company.
- CRM Customer Data, financial data from enterprise resource planning, monitoring tools' operational metrics etc.
- No need for rigid schemas
- Can add onto or modify system without affecting others. A boon for scalability
- Improved Interoperability

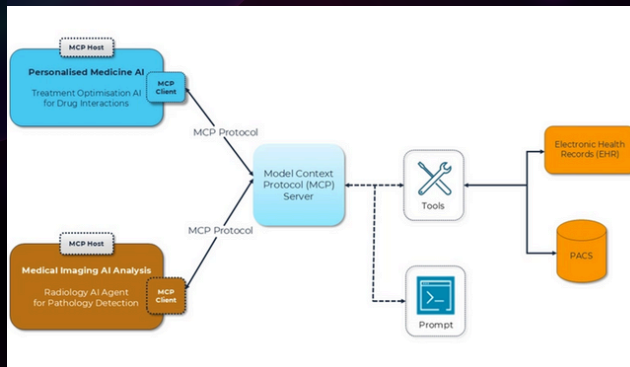


IN ENTERPRISE PART 2

- LLMs co-operational via MCP interfacing
- “Guide” for AI assistants
- Low latency in enterprise systems
- Fault-tolerance



SOME EXAMPLES



Customer Support Chatbots

Chatbots draw from previous chats, account settings/permissions so subsequent queries do not repeat unnecessary authentication steps and instead provide assistance relevant to user's current circumstances.

Medical Diagnostics

MCP gathers patient's past medical history, use of medication, symptoms, current condition to compile an information summary to the doctor



CONSIDERATIONS FOR IMPLEMENTATION

Support

Organisations must be informed what level of support for MCP third party services have.

Optimisation

Potential need to optimise performance

Mindful of Infrastructure

Companies must be wary of how data policies, security implementations and network resources affect implementation

Training

Potential need to retrain staff to work with MCP. Need to modify workflows?



PROS AND CONS

The Model Context Protocol (MCP) is a significant step forward in AI agent technology. However, it must be carefully evaluated, as with any developing normal . In order to determine its actual preparedness, we need to consider both its strong points and the significant obstacles it still faces.



PROS & CONS



Security & Trust

MCP server offers a structure way to handle connections , which can bed used to limit a sever server access. However it lacks the inherent security enforcement mechanisms. It is vulnerable to severe threats.

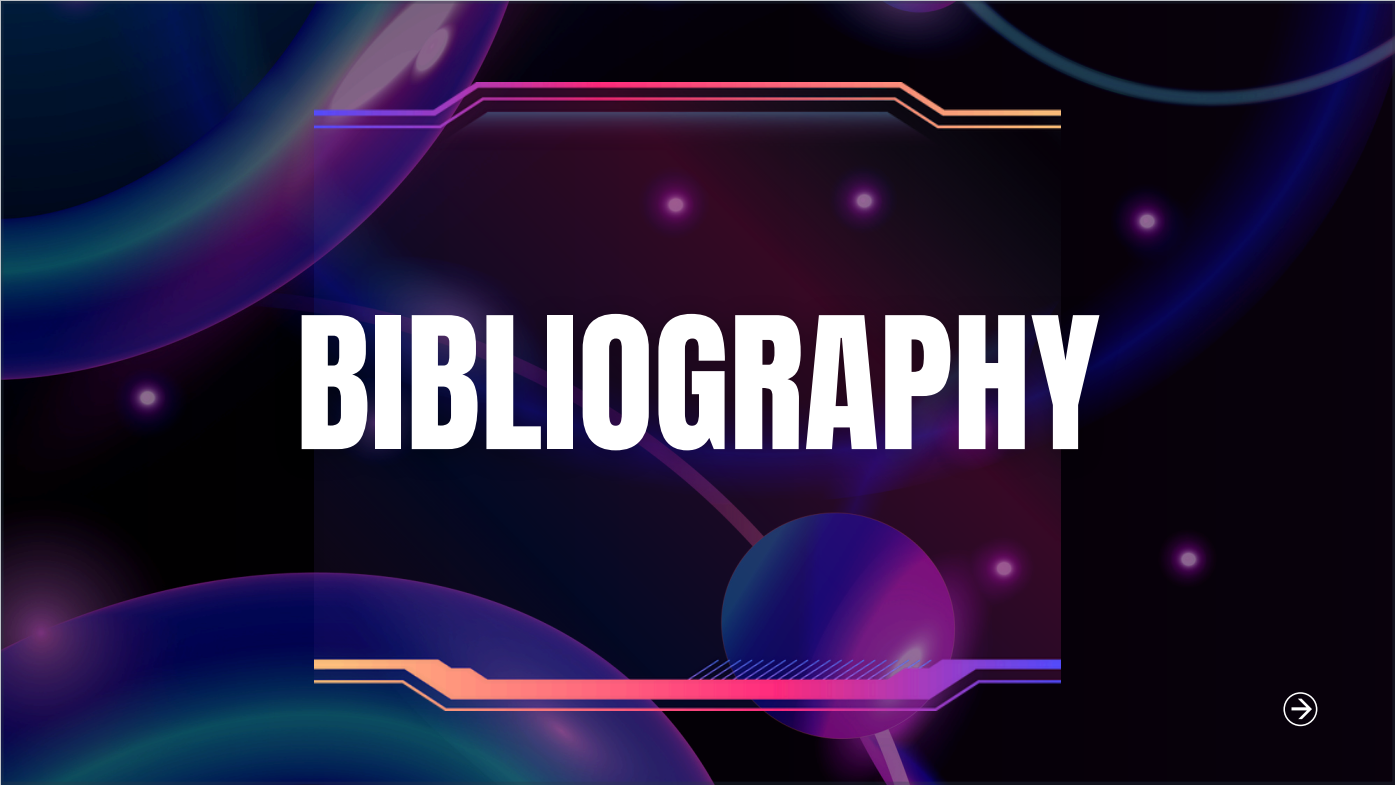
Data Handling and corrections

MCP has a advantage of being able to find and use technologies for data sharing in a systemaic ways .However , the protocol will undercuts this reliability by failing to enforce explicit criteria for faulty report , resulting in inconsistent and unreliable server feedback.

Future Visibility and Risk Investment

Although the growth of the MCP's leadership shows a dedication to stability and open standards. The unpredictability of the AI field constantly threatens this development, meaning that a younger, rival protocol might easily surpass today's investment.



The background of the page is a dark, futuristic space-themed illustration. It features glowing blue and purple nebulae, several bright purple stars, and a central blue sphere. Overlaid on this are stylized, glowing orange and blue lines that resemble circuitry or orbital paths. The word "BIBLIOGRAPHY" is prominently displayed in the center in a large, white, bold, sans-serif font.

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QUESTIONS?



THANK YOU!

